HANZ OFOSUHENE-SINTIM

Engineering Student (M1) — Electronics & Embedded Systems | Seeking a 3–4 Month Internship in Embedded C / Assembly, Testing & Analysis (April – July/August 2027)

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PROFILE

M1 Engineering student in Electronics and Embedded Systems at JUNIA ISEN Lille, concurrently completing my 4th-year BSc in Computer Engineering at KNUST (Ghana). Skilled in low-level programming (Embedded C, C/C++17, Assembly), microcontroller firmware (STM32, ESP32, MSP430), and systematic hardware-software testing, debugging, and analysis. Rigorous and autonomous, with proven executive leadership representing 800+ engineering students. Seeking a 3 to 4-month internship starting in April 2027 to contribute to embedded firmware development, testing, and system validation.

EDUCATION

JUNIA ISEN — Grande École d'Ingénieurs | Lille, France **2025 – Present (Expected 2028)**
Master's Degree in Engineering (M1) — Specialisation: Electronics and Embedded Systems

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (KNUST) | Kumasi, Ghana **2023 – June 2027 (In progress)**
Bachelor of Science (BSc) in Computer Engineering — 4th Year completed concurrently with JUNIA ISEN Master's

TECHNICAL PROJECTS

Autonomous Street-Cleaning Robot — JUNIA ISEN Lille (7-Member M1 R&D Team) **2026 – Present (Ongoing)**
- Developed real-time Embedded C firmware on an STM32 microcontroller for PWM H-bridge motor control, paired with ESP32 wireless telemetry.
- Conducted hardware-software integration testing, UART serial frame analysis (AI camera), and closed-loop control verification.

Digital Thermometer & Signal Acquisition Chain — MSP430 Microcontroller (C & Assembly) **2025**
- Designed and programmed low-level firmware in C and Assembly on a TI MSP430 microcontroller for precision temperature measurement.
- Analyzed and validated the complete signal processing chain from analog sensor capture and ADC conversion to calibrated digital readout.

Automated Water Level Control System — Arduino Uno & Ultrasonic Sensors **2024**
- Programmed deterministic microcontroller logic in C/C++ for automatic pump actuation governed by calibrated level thresholds.
- Performed sensor accuracy analysis and threshold hysteresis testing to ensure reliable closed-loop fluid regulation.

E.Leclerc Self-Checkout Terminal & Online Sales Architecture — C++17, OOP, STL **2024 – 2026**
- Architected a zero-dependency C++17 transaction engine evolving an online sales server architecture into an interactive self-checkout terminal.
- Executed boundary and unit testing on runtime polymorphic (dynamic_cast) weight verification and a 12-denomination Euro change algorithm.

PROFESSIONAL EXPERIENCE

Department President — ACES, KNUST (Association of Computer Engineering Students) **Aug 2025 – May 2026 (10 months)**
Higher Education Sector | Executive Board | Kumasi, Ghana
- Directed executive governance, budget management, and multidisciplinary teams representing 800+ Computer Engineering students.
- Spearheaded technical and logistical execution of CODEFEST 2026, coordinating industry sponsors, speakers, and technical judging panels.

Software Development & Testing Intern — SPAGARD TECHNOLOGIES **Oct – Dec 2023 (3 months)**
IT & Software Services Sector | Backend & Quality Assurance Unit | Ghana
- Developed server-side software components for internal enterprise management systems within a structured engineering workflow.
- Structured relational databases and executed systematic code debugging, root-cause analysis, and interface testing prior to release.

SKILLS

- **Low-Level & Programming:** Embedded C, C, C++17, Assembly (MSP430), Python 3, SQL, HTML/CSS, JavaScript
- **Embedded Systems & Hardware:** STM32, ESP32, MSP430, Arduino Uno, UART, PWM, Sensors, KiCad (PCB), Semiconductor Physics
- **Testing, Analysis & Tools:** Hardware-Software Debugging (GDB), Signal Chain Validation, Unit & Interface Testing, Linux, Git
- **Interpersonal (Soft Skills):** Team Leadership, Analytical Problem-Solving, Rigor, Autonomy, Cross-Cultural Communication

LANGUAGES & INTERESTS

- **Languages:** English: C2 (Native Speaker — Ghanaian by birth, full schooling & BSc in Ghana) | French: A2 (Elementary — Academic immersion in Lille, Duolingo Score: 21)
- **Interests & Extracurriculars:** Reading world history, exploring how physical/engineering systems work, Organizing Lead at GHAFRA Nord (Lille)